BTS Mobility Data Analysis Using Sequential and Distributed Processing

Module
5004CMD Data Science

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Introduction

https://github.com/Zyfiury/5004CMD-Data-Sciense

Though often unseen, mobility datasets capture how large groups move across distances and moments. These records help decode actual travel needs instead of assumed patterns. Where roads strain or emissions rise, such data reveal pressure points clearly. Real-world movement becomes a reference when cities rezone, tolls shift, or access reshapes behavior. Rather than predicting, they show what already happens - trip after trip.

Close distances mirror everyday routines: nearby zones show neighborhood trips, one after another. In between, moderate spans tie to work commutes and reaching public facilities. Farther reaches link areas across regions. Looking at these layers shows movement patterns scaled by distance, layer by layer. National trends unfold not as one number, but as shifts across ranges.

Despite regression connecting aligned daily sequences through bands, it captures joint shifts while offering basic validation via R squared and RMSE measured in actual trip counts. While Pandas serves smaller tasks well, shifting to Dask tackles larger loads - sequential runs prove easier to verify; parallel processing cuts runtime only if chunks outweigh overhead from coordination, data transfer, and task setup.

One goal involves summarizing weekly nationwide figures for people staying home alongside those not remaining at home. Weekly patterns in movement get condensed into clear summaries through separate tracking of these two groups. Another target focuses on measuring travel intensity across distances, using thresholds to highlight when mid-range journeys surpass ten million trips. This detection process marks specific days meeting that volume condition during analysis. A third task fits a straight-line model linking short-distance trips - specifically those between one and twenty-five miles - with five-to-ten-mile trip counts from combined national data points. Model accuracy appears through reported values for root mean square error along with coefficient of determination. The last aim compares processing duration for an identical analytical sequence run first through standard Pandas operations then repeated using Dask set up with both ten and twenty computing workers.

Datasets

Inside Trips_by_Distance.csv are entries marked by Level, Date, and Week, followed by figures showing how many people stayed home versus those who did not. This file also tracks overall travel numbers alongside grouped ranges like trips between five and ten or ten to twenty-five. Meanwhile, Trips_Full Data.csv logs nationwide daily movement across categories ranging from under one mile up to more than five hundred miles. It includes matching totals for both residents and total journeys, supporting broader analysis when combined with the other dataset. Though separate, each contributes distinct pieces needed to understand mobility patterns over time.

Beginning with national filtering ensures consistency within one region over time. When datetime values convert properly, sorting happens correctly - also allowing precise joins using date fields. Instead of letting duplicates skew results, removing them prevents exaggerated averages and faulty frequency counts. After keeping only National entries, extra location-based columns get discarded; this streamlines structure while minimizing type detection errors during parallel CSV processing.

Methodology

Pandas sets the starting point - frames operate predictably within a single process, handle CSVs and dates reliably, while built-in plotting supports consistent visuals. These features enable repeatable assignments plus benchmark results others later replicate at scale.

Through frame partitioning, Dask runs task graphs across a local cluster, processing reads and reductions in parallel while preserving the original analysis logic - this setup allows timing comparisons under identical computational workflows. Though structured differently behind the scenes, the sequence of operations stays unchanged, making runtime differences attributable to execution alone. By design, it mirrors standard procedures without modifying core methodology, ensuring fairness in performance measurement. Instead of rewriting analytics, it scales existing patterns through distributed mechanics. Even when spread across nodes, each step follows the intended path, just faster. Timing reflects infrastructure efficiency, not changes in reasoning or transformation rules.

One way to simplify daily fluctuations is grouping Trips_by_Distance.csv entries by week, focusing on people staying home versus those moving. Though it reduces noise, the weekly view keeps broader patterns clear in visual summaries like bar graphs. Instead of tracking individual days, this method highlights trends that shift gradually. When examining distance, analysts take mean values from National rows in Trips_Full Data.csv across various mile ranges. This averaging reveals how trip intensity changes with distance over time. By doing so, shifts in dominant travel behaviors become easier to identify within the overall U.S. pattern. The approach allows comparison between short and long trips without getting lost in daily variation.

Even when trip counts range from ten to twenty-five, thresholds single out days where activity passes ten million trips - either in that band or fifty to one hundred - to mark intense travel periods for clearer visual contrast. On matching dates within shared national coverage, regression links shorter journeys (1–25 miles) pulled from Trips_Full Data.csv with mid-range volume (5–10 trips) taken from Trips_by_Distance.csv to examine how movements align during the same day across differing spans.

Typically, RMSE reflects average differences in predicted versus actual trips. What sets R squared apart is its comparison against a baseline that uses only the mean - this distinction clarifies how much variation a model actually accounts for. While one measures magnitude of mistakes, the other highlights explanatory strength, helping viewers separate overall fit from raw precision.Exploratory Data Analysis

Weekly aggregation matches common reporting cadences and improves cross-month comparability. Population staying at home signals reduced out-of-home time and is sensitive to gradual shifts in routine participation. Population not staying at home captures realised travel on the same days and completes the national activity budget implied by the two population fields. Distance comparisons matter because headline totals can conceal concentration in short bands that load local networks and environmental exposures. Short-distance dominance is expected when daily routines remain localised even if occasional long trips are socially salient.

Figures 1–3 present weekly home and non-home averages and cross-sectional distance means before thresholding and regression.

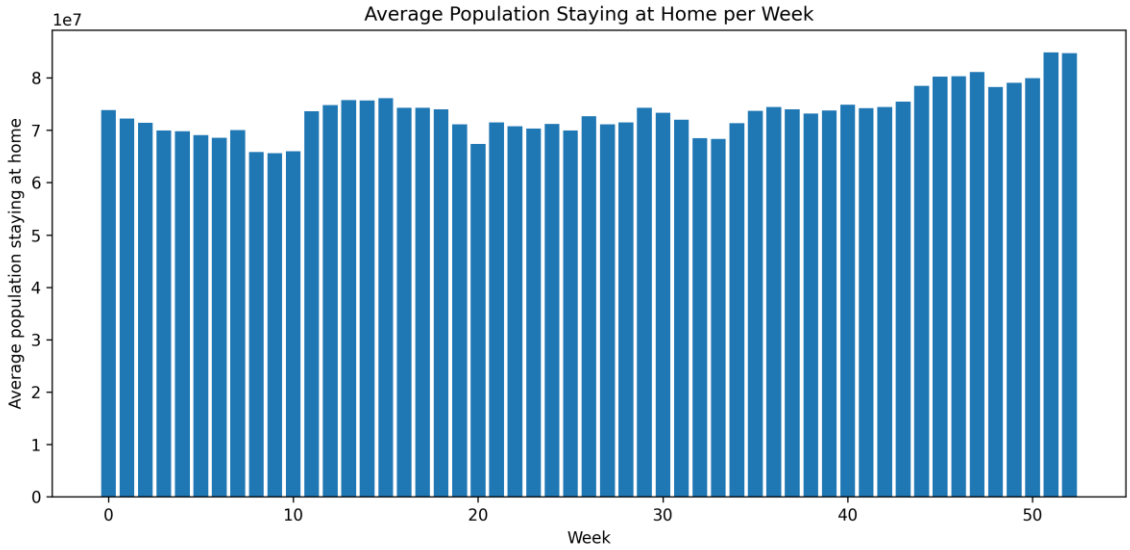

Figure 1. Average population staying at home per week

Figure 1 shows weekly mean Population Staying at Home varying gradually without sharp single-week breaks, consistent with slow routine adjustment rather than episodic shocks at this resolution. Late-sample increases, where visible, imply modestly higher average home time and should be read with Figure 2 to infer net mobility change.

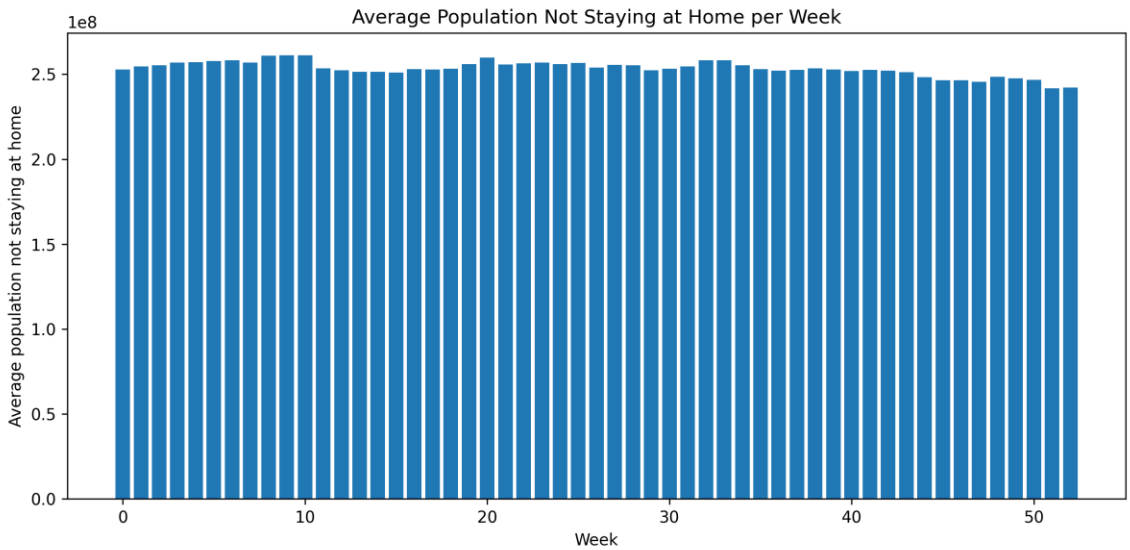

Figure 2. Average population not staying at home per week

Figure 2 shows Population Not Staying at Home persistently large and comparatively stable, indicating structurally high national circulation with only modest week-to-week variation around a high mean. Together with Figure 1, the series suggests gradual co-movement within a stable aggregate envelope.

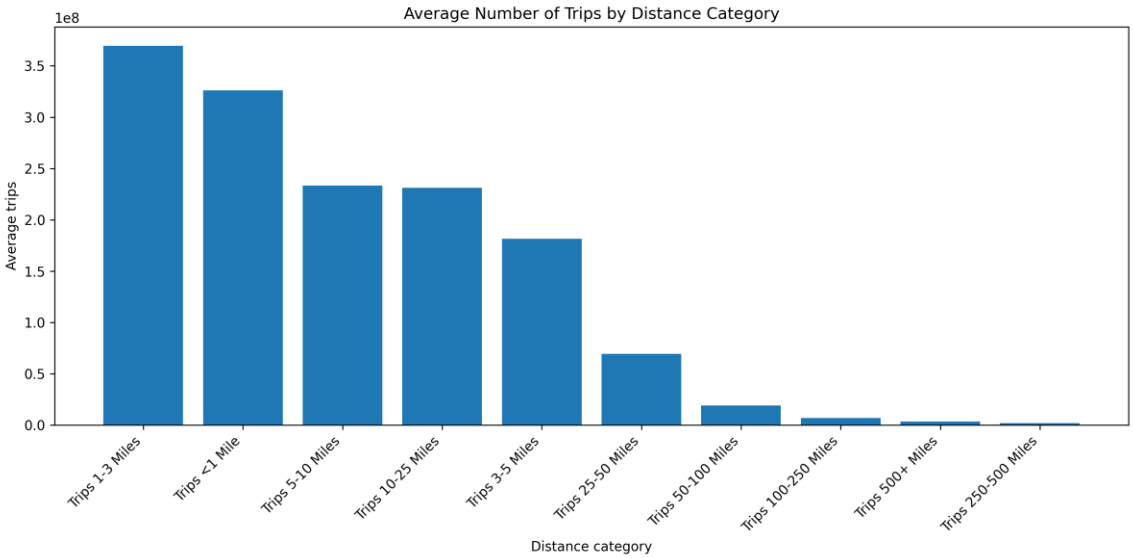

Figure 3. Average trips across distance categories

Figure 3 places the highest means in short bands, led by Trips 1-3 Miles, consistent with local access trips and short segments within wider journeys. This implies commuting systems still generate massive intra-local circulation, so junction reliability, local transit

frequency, and active-travel safety remain high-leverage even when longer commutes exist.

Threshold Analysis

Ten million trips mark an extreme national daily count that is simple to communicate across bins. Ten-to-twenty-five-mile movement captures intra-regional circulation between suburbs and centres, distinct from purely local or rare long-haul travel. Frequent medium-band exceedance relative to a longer band suggests repeatable medium-range travel drives many high-volume days. Figure 4 shows temporal clustering of exceedances.

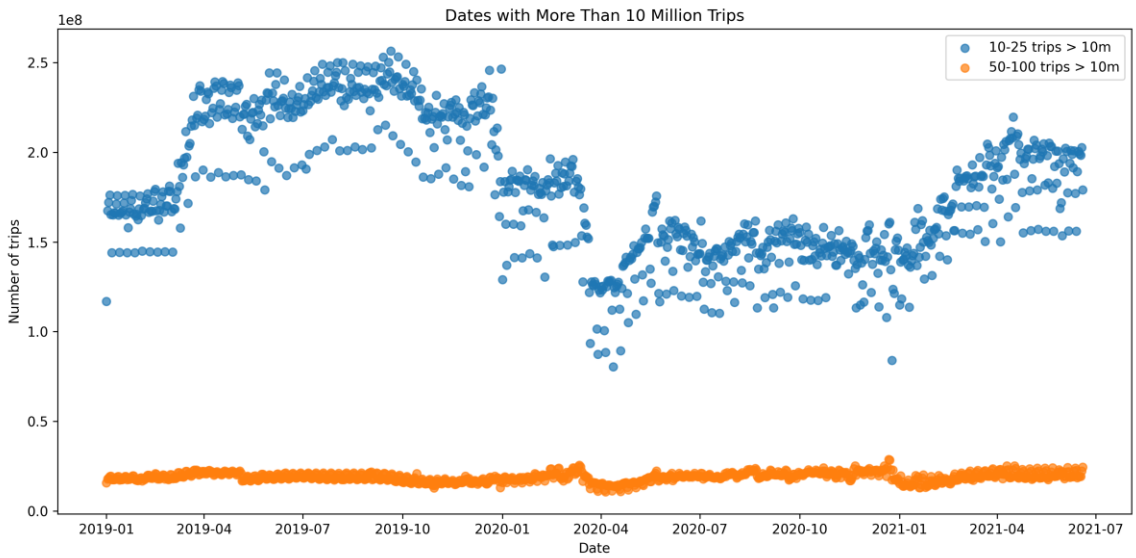

Figure 4. Trips above ten million threshold over time

Regression Model

Trips 1-25 Miles concentrates short-to-medium national volume into one predictor aligned with exploratory distance structure. Number of Trips 5-10 targets an adjacent operational bin beyond the shortest hops, merged on Date for national rows. A positive slope indicates same-day co-movement consistent with shared activity budgets. Coefficient, intercept, RMSE, and R2 are reported in Model Evaluation.

Figure 5 plots observations with the ordinary least squares line. Linear assumptions include an approximately linear conditional mean and limited outlier leverage; violations can mislead even when fit appears strong.

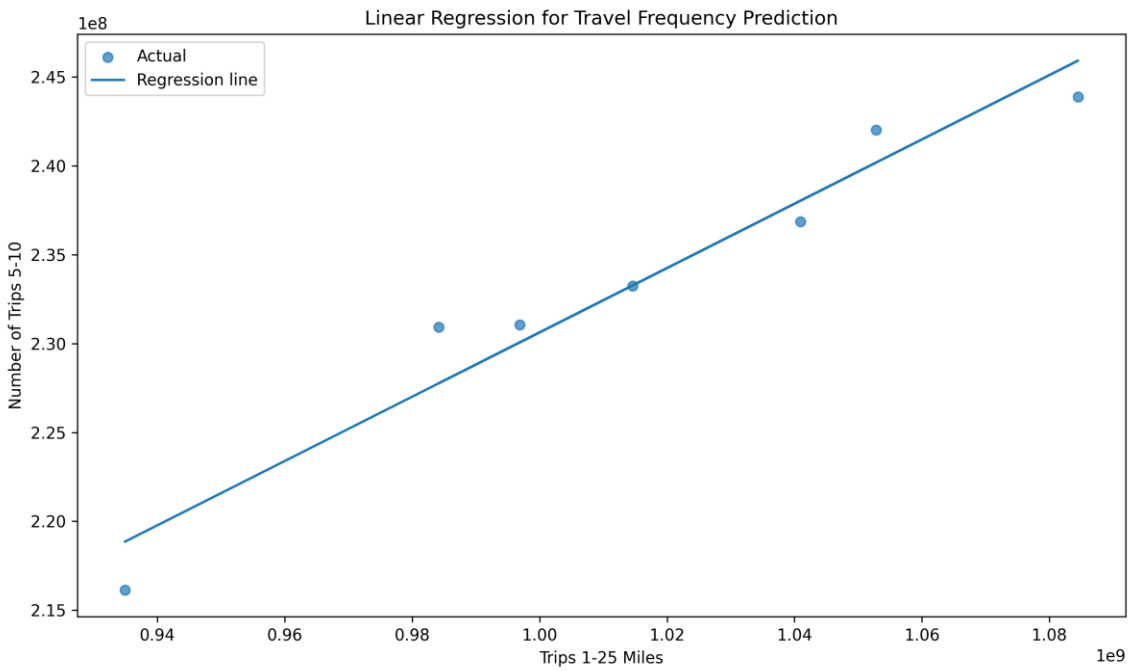

Figure 5. Linear regression between distance categories

Model Evaluation

Ordinary least squares on nationally merged daily observations yields the metrics below in original trip-count units.

R2 equals 0.946

RMSE equals 1981454.86

Coefficient equals 0.180985

Intercept equals 49643925.69

R2 equals 0.946 implies the linear predictor explains most day-to-day variation in Number of Trips 5-10 conditional on Trips 1-25 Miles in the merged window, signalling a tight linear association for descriptive reporting. RMSE equals 1981454.86 remains large in absolute terms because national daily counts are enormous, so even proportionally small errors translate into large trip-count deviations. Coefficient and intercept should be read jointly: the slope describes incremental co-movement between bands, while the intercept positions the line. Limitations include strict linearity, omitted weekly or shock structure, and sensitivity to merge coverage across Trips_Full Data.csv and Trips_by_Distance.csv.

Travellers by Distance Band

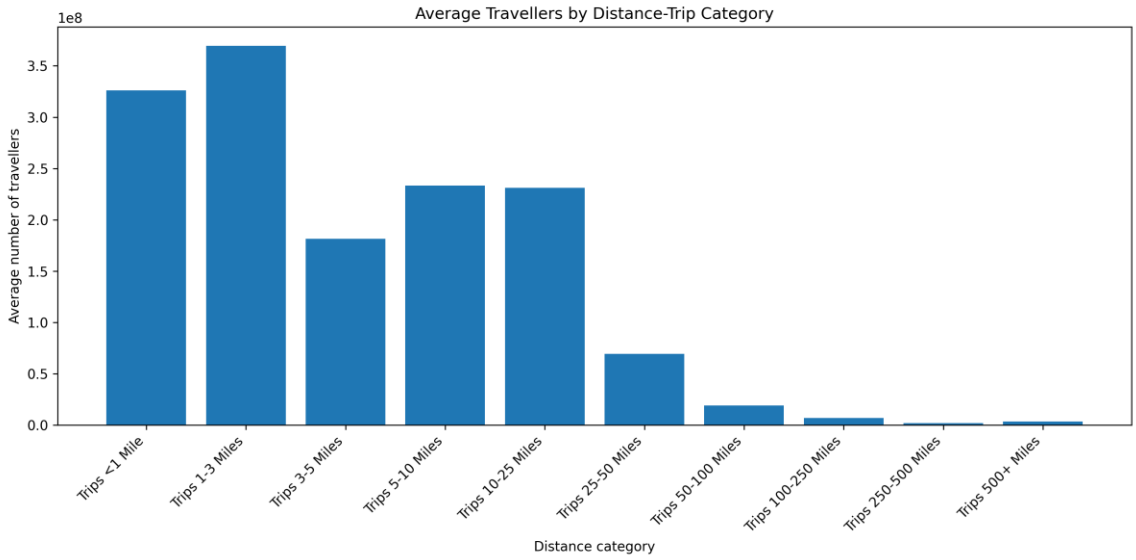

Figure 6. Distribution of travellers by distance band

Figure 6 shows declining average travellers as distance increases, concentrating routine intensity in short bands and leaving long bands with comparatively small averages. That skew implies infrastructure stress and local air-quality exposures track short-range circulation most closely. Planning therefore prioritises junction capacity, local transit headways, and walking and cycling safety, while longer corridors remain important for connectivity even though they carry a smaller share of average daily mass in these summaries.

Parallel Processing Comparison

Processing mode	Time (seconds)
Serial processing	3.99
Parallel processing with ten workers	9.06
Parallel processing with twenty workers	9.58

The benchmark asks whether partitioned Dask execution accelerates the replicated pipeline relative to serial Pandas on the same logical steps. Sequential runs minimise coordination and keep data movement local to one process. Distributed runs add a scheduler, worker processes, serialization, and partitioned CSV reads—fixed and variable overhead that can dominate elapsed time when partitions are small or when results must be collected frequently to the client.

Performance scales with size: larger data amortises fixed cluster costs across more bytes and tasks, whereas smaller extracts spend disproportionate time on graph construction

and startup. Dask is more compelling for large, repeated partitioned pipelines and when out-of-core execution avoids memory ceilings. In the recorded coursework run, serial time was 3.99 seconds versus 9.06 seconds with ten workers and 9.58 seconds with twenty workers, illustrating coordination overhead under the tested configuration and moderate data scale.

Discussion

Short-band dominance structures daily activity around local access and repeated short hops, while medium-distance travel sustains regional connectivity and can spike above the ten-million threshold on busy days. Regression ties Trips 1-25 Miles to Number of Trips 5-10, supporting simple monitoring, nowcast-style checks, and communication of co-movement to non-technical stakeholders. Timing results caution that parallelism is not automatically faster; tool choice should follow data scale, task granularity, and how often results must be consolidated.

Limitations

Hidden differences between regions vanish when data are pooled nationally. Because seasons overlap with holidays, policies, and long-term shifts, patterns can blur together. Trends that assume straight-line change might ignore sudden breaks or shifting phases in movement behavior. Timing analyses using modest sample sizes gain little from parallel processing since overhead eats into gains. Missing factors - like fuel prices, climate swings, or job market disruptions - weaken cause-effect claims. Unaccounted forces still shape what remains after modeling.

Conclusion

Trips_by_Distance.csv and Trips_Full Data.csv jointly portray weekly home and non-home behaviour, short-band dominance, and medium-distance high-volume days. Regression links Trips 1-25 Miles to Number of Trips 5-10 with R2 equal to 0.946 and RMSE equal to 1981454.86. Serial Pandas outpaced Dask with ten and twenty workers in the recorded benchmark, consistent with coordination overhead. Future work may add geography, seasonality, nonlinear models, larger extracts to retest parallelism, and policy covariates.

Appendix

Pseudocode

Appendix A.2 Pseudocode

Algorithm: BTS Mobility Analysis Workflow
Input:
- Trips_by_Distance.csv
- Trips_Full Data.csv

Output:
- Weekly home and non-home analysis
- Threshold comparison for trip ranges
- Regression model metrics
- Distance-based plots
- Serial and parallel timing comparison

Begin
1. Load both datasets.
2. Convert Date columns to date format.
3. Filter both datasets to National level only.
4. Remove unused location columns from Trips_by_Distance data.
5. Remove duplicate rows.
6. Group Trips_by_Distance data by Week.
7. Compute the average values for:
   7.1 Population Staying at Home
   7.2 Population Not Staying at Home
8. Plot weekly bar charts for home and non-home populations.
9. From Trips_by_Distance data, select rows where:
   9.1 Number of Trips 10-25 > 10,000,000
   9.2 Number of Trips 50-100 > 10,000,000
10. Count matching dates for each condition.
11. Plot a scatter chart to compare the two threshold groups.
12. Match date ranges between the two datasets.
13. Build a linear regression model using:
    13.1 Predictor: Trips 1-25 Miles
    13.2 Response: Number of Trips 5-10
14. Generate model predictions.
15. Calculate RMSE and R2.
16. Plot the regression scatter graph and fitted line.
17. Compute average travellers by distance category from Trips_Full Data.
18. Plot the distance-trip bar chart.

19. Run the workflow in serial mode.
20. Record the serial execution time.
21. Run the workflow in parallel mode with 10 workers.
22. Record the 10-worker execution time.
23. Run the workflow in parallel mode with 20 workers.
24. Record the 20-worker execution time.
25. Compare all recorded timings in a summary table.
26. Save charts and timing outputs for the report.
End

Flowchart description

Appendix A.1 Flowchart Draft

Start
↓
Load Trips_by_Distance.csv and Trips_Full Data.csv
↓
Convert Date fields
↓
Filter National level rows
↓
Clean and remove irrelevant columns
↓
Question 1:
Compute weekly home and non-home averages
↓
Create weekly bar charts
↓
Question 2:
Filter dates above 10,000,000 for 10-25 and 50-100 trip ranges
↓
Create threshold comparison scatter plot
↓
Question 3:
Align dates and train linear regression model
↓
Evaluate RMSE and R2
↓
Create regression figure
↓
Question 4:

Compute average travellers by distance category
↓
Create distance-trip bar chart
↓
Run serial workflow
↓
Record time
↓
Run parallel workflow with 10 workers
↓
Record time
↓
Run parallel workflow with 20 workers
↓
Record time
↓
Create timing comparison table
↓
Insert outputs into scientific paper
↓
End

GitHub repository

Full reproducible code, datasets, figures, and workflow:

https://github.com/Zyfiury/5004CMD-Data-Sciense